CS102 – Assignment 2 Lab Extension

Group 1 (Students with Even Student IDs): Word Generation

Objective: To generate a single word of a specified length, starting with a letter chosen probabilistically, and then building the rest of the word based on letter-following frequencies.

Steps:

1. **Prompt for Length:** Your program should ask the user to input a desired length for the word they want to generate (e.g., 10).
2. **Determine Initial Letter:** Using your **WordList** (containing unique words), analyze the frequency of each first letter among these unique words. Implement a mechanism to select the starting letter for your generated word based on these calculated probabilities.
3. **Generate Subsequent Letters:** For each letter after the initial one, you will use the "Proceeding letter ratio" data that you computed for the third analysis method in the original assignment. Continuously select the next letter using weighted random selection based on these ratios until the generated word reaches the user-specified length.

Note: You should implement the weighted random selection manually using Math.random() or Random.nextDouble(). Do **not** use any external libraries for probability-based selection.

Group 2 (Students with Odd Student IDs): Text Generation

Objective: To generate a sequence of words (a text) of a specified length, starting with a user-provided word and then building the text based on word-to-word transition probabilities. For this calculate the proceeding word ratios for every word.

Steps:

1. **Prompt for Initial Word and Length:** Your program should ask the user for a starting word and the desired number of words for the generated text.
2. **Build a Word Transition Model:** You will need to process the *entire* original text file (considering all words, not just unique ones) to create a model that captures how often one word follows another. This model should allow you to determine the probability of any given word being followed by another specific word.
3. **Generate Text:** Begin with the initial word provided by the user. Then, using your word transition model, repeatedly select the next word in the sequence through weighted random selection. Continue this process until the generated text contains the user-specified number of words. Consider how to handle situations where a chosen word has no recorded followers in your model.

Note: You should implement the weighted random selection manually using Math.random() or Random.nextDouble(). Do **not** use any external libraries for probability-based selection.